

Chapter 3

Homework 5 due: Thursday 10th April, 2014

All problems without a ★ from section 4.1; exercises 3.14-3.19.

Exercise 3.14. ★

Exercise 3.15. Give two examples of elements in $\mathbb{Z}/4\mathbb{Z} = \mathbb{Z}/\langle 4 \rangle$, and add and multiply them.

Proof. Possible elements are $0 + \langle 4 \rangle = \langle 4 \rangle$, $1 + \langle 4 \rangle$, $2 + \langle 4 \rangle$, $3 + \langle 4 \rangle$.

$$\bullet a + \langle 4 \rangle = \{ \dots, a - 4, 0 + a, 4 + a, 8 + a, \dots \}$$

□

Take $1 + \langle 4 \rangle$ and $2 + \langle 4 \rangle$.

$$(1 + \langle 4 \rangle) \oplus (2 + \langle 4 \rangle) = (1 + 2) + \langle 4 \rangle = 3 + \langle 4 \rangle = 1 + \langle 4 \rangle$$

$$(1 + \langle 4 \rangle) \otimes (2 + \langle 4 \rangle) = 2 + \langle 4 \rangle$$

Exercise 3.16. Find all the distinct cosets of $\mathbb{Z}/4\mathbb{Z}$.

Proof. Recall that $\mathbb{Z}/\langle 4 \rangle \cong \mathbb{Z}_4$ so it has 4 elements. From above they are $0 + \langle 4 \rangle = \langle 4 \rangle$, $1 + \langle 4 \rangle$, $2 + \langle 4 \rangle$, $3 + \langle 4 \rangle$.

To see that they are distinct note that $4 \nmid a - b$ for any $0 \leq a, b \leq 3$ since then $a - b < 4$. □

Exercise 3.17. Let $S = \{a + bi : b \in 2\mathbb{Z}\}$ (b is even and $i = \sqrt{-1}$). Prove that S is a subring of $\mathbb{Z}[i]$ but that S is not an ideal of $\mathbb{Z}[i]$.

Proof. Let $a + 2ni, a' + 2mi \in S$ for some $a, a', n, m \in \mathbb{Z}$. Using the subring test,

$$(a + 2ni) - (a' + 2mi) = (a - a') + 2(n - m)i \in S$$

and

$$(a + 2ni)(a' + 2mi) = aa' + 2ami + 2ani - 2nm = (aa' - 2nm) + 2(m + n)i \in S$$

so S is a ring.

To show that S is not an ideal in $\mathbb{Z}[x]$ consider $1 + 2i \in S$ and $1 + i \in \mathbb{Z}[x]$.

$$(1 + 2i)(1 + i) = 2 + 2i + i - 2 = 0 + 3i \notin S.$$

□

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Exercise 3.18. Let $S = \{a + bi : b \in 2\mathbb{Z}\}$ (b is even and $i = \sqrt{-1}$). Prove that $\mathbb{Z}[i]/S$ is not a ring.

Proof. By exercise 3.4 S is not an ideal. So by theorem 4.3 $\mathbb{Z}[i]/S$ is not a ring. $\square$

Exercise 3.19. Prove that $\mathbb{R}[x]/\langle x^2 + 1 \rangle$ is a ring. (We know that the ring of polynomials with integer coefficients, $\mathbb{R}[x]$, is a ring).

Proof. By theorem 3.4 we need only show that $\langle x^2 + 1 \rangle$ is an ideal. If $g_1(x), g_2(x) \in \langle x^2 + 1 \rangle$ then $g_1(x) = (x^2 + 1)f_1(x)$ and $g_2(x) = (x^2 + 1)f_2(x)$ for some $f_1(x), f_2(x) \in \mathbb{R}[x]$. Using the ideal test,

$$\begin{aligned} g_1(x) - g_2(x) &= (x^2 + 1)f_1(x) - (x^2 + 1)f_2(x) \\ &= (x^2 + 1)(f_1(x) - f_2(x)) \in \langle x^2 + 1 \rangle \end{aligned}$$

and for any $h(x) \in \mathbb{R}[x]$,

$$h(x)g_1(x) = g_1(x)h(x) = h(x)((x^2 + 1)f_1(x)) = h(x)f_1(x)(x^2 + 1) \in \langle x^2 + 1 \rangle$$

so $\langle x^2 + 1 \rangle$ is an ideal which implies that $\mathbb{R}[x]/\langle x^2 + 1 \rangle$ is a ring by the theorem. $\square$

Exercise 3.20. Give two examples of elements in $\mathbb{Z}[i]/\langle 2 - i \rangle$, and add and multiply them.

Proof. $1 + \langle 2 - i \rangle$ and $(2 + i) + \langle 2 - i \rangle$ are two elements in the factor ring $\mathbb{Z}[i]/\langle 2 - i \rangle$.

$$[1 + \langle 2 - i \rangle] \oplus [(2 + i) + \langle 2 - i \rangle] = (3 + i) + \langle 2 - i \rangle$$

and

$$[1 + \langle 2 - i \rangle] [(2 + i) + \langle 2 - i \rangle] = (2 + i) + \langle 2 - i \rangle.$$

$\square$

- Note that $2 - i \equiv 0 \pmod{\langle 2 - i \rangle}$ so then $2 \equiv i$, and $(2 + i) + \langle 2 - i \rangle \equiv 4 + \langle 2 - i \rangle$. We can easily check this: $(2 + i) - 4 = -2 + i = -1(2 - i) \in \langle 2 - i \rangle$.

Exercise 3.21. Find all the distinct cosets of $\mathbb{Z}[i]/\langle 2 - i \rangle$.

Proof. Let $\langle 2 - i \rangle \in \mathbb{Z}[i]$. $\mathbb{Z}[i] = \{x + yi : x, y \in \mathbb{Z}\}$, called the *Gaussian integers*. For any two Gaussian integers z and z' we have $(2 - i)z - (2 - i)z' = (2 - i)(z - z') \in \langle 2 - i \rangle$, and $z_1(z_2(2 - i)) = (z_2(2 - i))z_1 = z_1z_2(2 - i) \in \langle 2 - i \rangle$. So $\langle 2 - i \rangle$ is an ideal of $\mathbb{Z}[i]$.

By the theorem, $\mathbb{Z}[i]/\langle 2 - i \rangle$ is a ring with elements (cosets) of the form $x + yi + \langle 2 - i \rangle$ for $x, y \in \mathbb{Z}$. When dealing with cosets in $\mathbb{Z}[i]/\langle 2 - i \rangle$ we can treat $2 - i$ as 0 since $2 - i + \langle 2 - i \rangle = 0 + \langle 2 - i \rangle$. This allows us to simplify the set of *distinct* cosets. For example, since $2 - i \equiv 0$ we also know that $2 \equiv i$ since $2 - i + i = 2$. So then

$$6 + 3i + \langle 2 - i \rangle = 6 + 3(2) + \langle 2 - i \rangle = 12 + \langle 2 - i \rangle$$

$2 \equiv i \Rightarrow 4 \equiv -1$ and $4 + 1 = 5 \equiv 0$. So we should expect there to be 5 distinct cosets, and

$$\mathbb{Z}[i]/\langle 2 - i \rangle = \{0 + \langle 2 - i \rangle, 1 + \langle 2 - i \rangle, \dots, 4 + \langle 2 - i \rangle\}$$

To see that this is the case, note that any other coset will be some integer multiple of $1 + \langle 2 - i \rangle$. Since $5(1 + \langle 2 - i \rangle) = 0 + \langle 2 - i \rangle$, $5(1 + \langle 2 - i \rangle)$ has order 1 or 5. If it is 1, then $1 + \langle 2 - i \rangle = 0 + \langle 2 - i \rangle$ and $1 \in \langle 2 - i \rangle$ so $1 = (x + yi)(2 - i) = 2x - ix + 2yi + y = 2x + y + i(-x + 2y)$ for some integers x and y . Since $1 \in \mathbb{R}$ this implies that $1 = 2x + y$ and $0 = -x + 2y$. Solving this system gives $y = 1/5 \notin \mathbb{Z}$, a contradiction. So the order of $\mathbb{Z}[i]/\langle 2 - i \rangle$ is 5. From the operations we can also see that this ring is essentially the same as $\mathbb{Z}_5$, a field. $\square$